

State-Corrected GRPO: Recovering the Dropped State-Distribution Ratio in LLM Policy Optimization

Status. Working draft. Introduction deliberately omitted — this draft focuses on the core idea and current experimental evidence. All results are on **Qwen2.5-1.5B-Instruct**, MATH, 105 steps, group size $G = 8$, **ver1** framework.

1 · Motivation

Start on-policy. The vanilla policy-gradient (REINFORCE) objective is

$$\nabla J(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a | s) \cdot \hat{A}(s, a) \right].$$

Both the state distribution d^{π_θ} and the action distribution $\pi_\theta(\cdot | s)$ are under the *current* policy — the algorithm is statistically correct only when every rollout used to compute the gradient came from today’s π_θ . For LLMs this is prohibitive: rollouts cost far more than gradient steps, so we want to **reuse samples across updates** (multi-epoch PPO), collect them asynchronously (async RL frameworks), or continue using a rollout even after the policy has drifted.

Going off-policy forces an importance-sampling correction. If rollouts come from a behaviour policy $\pi_\beta \neq \pi_\theta$, the on-policy expectation can be rewritten, exactly, as

$$\nabla J(\theta) = \mathbb{E}_{s \sim d^{\pi_\beta}, a \sim \pi_\beta} \left[\underbrace{\frac{d^{\pi_\theta}(s)}{d^{\pi_\beta}(s)}}_{\text{state ratio}} \cdot \underbrace{\frac{\pi_\theta(a|s)}{\pi_\beta(a|s)}}_{\text{action ratio}} \cdot \nabla_\theta \log \pi_\theta(a | s) \cdot \hat{A}(s, a) \right].$$

Two IS correction terms appear — one for the state distribution, one for the action. Both are needed for the identity to hold.

The TRPO/PPO/GRPO lineage: keep one, drop one. The state ratio is notoriously hard to estimate in a general MDP (the numerator requires knowing d^{π_θ} , which has no closed form), so a long line of methods takes the same two-step shortcut:

1. **Drop the state ratio** — set $d^{\pi_\theta}(s)/d^{\pi_\beta}(s) \equiv 1$.
2. **Compensate by constraining the action ratio** close to 1 — via a KL trust region (TRPO), a clipped ratio (PPO/GRPO/DAPO), a geometric-mean sequence ratio (GSPO), a soft gate (SAPO), a direct divergence (DPPO), or a prefix-min surrogate (MinPRO).

The justification is always the same: **if $\pi_\theta(\cdot | s)$ stays close to $\pi_\beta(\cdot | s)$ at every state, then $d^{\pi_\theta} \approx d^{\pi_\beta}$** (a standard TRPO-style performance-difference bound), so the state-ratio-free surrogate is a good approximation of the true off-policy gradient. Every GRPO variant can be read as a different answer to "how do we keep the action ratio in check?" — but none touches the state ratio directly. This is the entire design space of the last decade of policy-gradient methods for language models: one dropped term, many action-side trust-region formulations.

The approximation is tight in the strict on-policy limit. It degrades under moderate off-policy drift — async rollouts, multi-epoch updates, long sequences — precisely the regimes modern LLM training is trending toward. And it is, structurally, a workaround: restricting the action ratio to avoid a state-ratio problem you’d rather not compute.

Key observation. For LLMs the state ratio is not intractable. Transitions are deterministic: $s_{t+1} = (q, o_{\leq t})$ is uniquely determined by (s_t, a_t) , so

$$d^\pi(s_t) = \prod_{j=1}^{t-1} \pi(o_j \mid q, o_{<j}) \quad \implies \quad \boxed{\frac{d^\pi(s_t)}{d^{\pi_\beta}(s_t)} = \prod_{j=1}^{t-1} r_j(\theta), \quad r_j(\theta) = \frac{\pi_\theta(o_j \mid q, o_{<j})}{\pi_\beta(o_j \mid q, o_{<j})}.}$$

This identity is **exact** — every r_j is already computed during training. The question shifts from "how hard should we clamp the action ratio to fake $d^\pi \approx d^{\pi_\beta}$?" to "**how do we directly use $\prod_{j<t} r_j$ without the classical exponential-variance explosion of trajectory IS?**" The rest of this paper answers the second question.

2 · The SC-GRPO Objective

Let r_t be the per-token action ratio and w_t an **exclusive state weight** depending only on the prefix $o_{<t}$ (various surrogates in §3):

$$\boxed{L_{\text{SC}}(\theta) = -\mathbb{E}_{o \sim \pi_\beta} \left[\sum_t w_t \cdot r_t \cdot \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \cdot m_t \right].}$$

Both w_t and r_t are detached; gradient flows only through $\log \pi_\theta$. The unified trust-region gate is

$$m_t = \mathbf{1}[w_t \in [w_{\min}, w_{\max}]] \cdot \mathbf{1}[r_t \in [1 - \epsilon, 1 + \epsilon]].$$

SC clamp and PPO clip are treated symmetrically as multiplicative gradient masks — we call this **Plan A**. It produces a single diagnostic (`joint_active_frac`) that faithfully reflects which tokens drive the gradient.

Implementation note. For state correction to be meaningful, `old_log_prob` must be the true rollout log-prob $\log \pi_\beta$, not a recomputed $\pi_{\theta_{\text{old}}}$. We obtain it via `verl`'s `bypass_mode`, which feeds vllm-side rollout log-probs through as old log-probs — saving one actor forward pass and guaranteeing π_β is exact.

3 · A Unified View of $w_{i,t}$

Strip every PPO-family method down to its policy-gradient shape. Ignoring trust-region gates (clip / clamp — they are gradient masks, not part of the off-policy IS structure), every method below can be written as a per-token objective of the form

$$L(\theta) = -\mathbb{E}_{\{o_i\} \sim \pi_\beta} \left[\sum_i \sum_{t=1}^{|o_i|} w_{i,t} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \log \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \right],$$

which gives the gradient

$$\nabla L(\theta) = -\mathbb{E} \left[\sum_{i,t} w_{i,t} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \nabla_\theta \log \pi_\theta(o_{i,t}) \right],$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t}) / \pi_\beta(o_{i,t} \mid q, o_{i,<t})$ is the **per-token action ratio** and $w_{i,t}$ is the **state weight**. The exact off-policy objective has $w_{i,t}^{\text{exact}} = \rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$. Every method differs from the exact case in exactly one place: what it puts in $w_{i,t}$.

3.1 GRPO variants, reduced to a common form

GRPO. The published objective (ignoring clip):

$$\mathcal{J}_{\text{GRPO}} = \mathbb{E} \left[\frac{1}{G} \sum_{g=1}^G \frac{1}{|o_i|} \sum_{t=1}^T r_{i,t} \hat{A}_{i,t} \right].$$

The $\frac{1}{G} \sum_{g=1}^G \frac{1}{|o_i|} \sum_{t=1}^T$ factor is a Monte-Carlo aggregation choice (estimator of the outer expectation), not a per-token weight specified by the algorithm. Dropping it into the outer expectation, the per-token shape is $r_{i,t} \hat{A}_{i,t} \cdot \log \pi_\theta$ — i.e., $w_{i,t} = 1$.

$$\boxed{w_{i,t}^{\text{GRPO}} = 1.}$$

DAPO. Same as GRPO except the aggregation is changed to **token-mean** (a correct MC estimator) and the clip becomes asymmetric. Neither changes $w_{i,t}$:

$$\mathcal{J}_{\text{DAPO}} = \mathbb{E} \left[\frac{1}{\sum_{i'} |o_{i'}|} \sum_{i,t} r_{i,t} \hat{A}_{i,t} \right] \implies \boxed{w_{i,t}^{\text{DAPO}} = 1.}$$

GSPO. Defines a sequence-level ratio $s_i = (\prod_{j=1}^{|o_i|} r_{i,j})^{1/|o_i|}$ and writes

$$\mathcal{J}_{\text{GSPO}} = \mathbb{E} \left[\frac{1}{G} \sum_i s_i \hat{A}_i \right].$$

The gradient is $\nabla s_i = s_i \cdot \frac{1}{|o_i|} \sum_t \nabla \log \pi_\theta(o_{i,t})$. Apply the identity $\nabla \log \pi_\theta = \frac{1}{r_{i,t}} \cdot r_{i,t} \nabla \log \pi_\theta$ once to align with the common form:

$$\nabla \mathcal{J}_{\text{GSPO}} = \mathbb{E} \left[\sum_{i,t} \frac{s_i}{|o_i| r_{i,t}} \cdot r_{i,t} \nabla \log \pi_\theta \cdot \hat{A}_i \right] \implies \boxed{w_{i,t}^{\text{GSPO}} = \frac{s_i}{|o_i| r_{i,t}}.}$$

s_i depends on the whole response including future tokens $o_{>t}$, so $w_{i,t}^{\text{GSPO}}$ is not state-measurable — it breaks the causal structure that a variance-reducing baseline requires (§3.4).

MinPRO. Defined directly as a per-token objective with $w_{i,t} = \min_{j < t} r_{i,j}$:

$$\mathcal{J}_{\text{MinPRO}} \propto \mathbb{E} \left[\sum_{i,t} \underline{\rho}_{i,t} \cdot r_{i,t} \hat{A}_{i,t} \right], \quad \underline{\rho}_{i,t} = \min_{j < t} r_{i,j} \implies \boxed{w_{i,t}^{\text{MinPRO}} = \min_{j < t} r_{i,j}.}$$

MinPRO is the first method in the table to put a real causal prefix statistic into $w_{i,t}$ — stable (non-cumulative), conservative (always $\leq \rho_{i,1:t-1}$).

3.2 SC-GRPO variants, same recipe

Our SC family replaces $w_{i,t} = 1$ by a prefix-based surrogate that targets $\rho_{i,1:t-1}$. The per-token objective is always

$$L_{\text{SC}}(\theta) = -\mathbb{E} \left[\sum_{i,t} w_{i,t}^{\text{SC}} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \log \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \right];$$

the four variants differ only in $w_{i,t}^{\text{SC}}$. To understand the trade-off, note that the exact weight $\rho_{i,1:t-1}$ has

$$\mathbb{E}_{\pi_\beta}[\log \rho_{1:t-1}] \approx -(t-1)\bar{D}_{\text{KL}}, \quad \text{Var}_{\pi_\beta}[\log \rho_{1:t-1}] \lesssim (t-1)\sigma^2, \quad \text{Var}_{\pi_\beta}[\rho_{1:t-1}] \gtrsim e^{(t-1)\sigma^2} - 1,$$

i.e. the log-weight drifts linearly negative (Jensen) while the weight itself has exponential variance. The four variants trade bias for variance differently:

Variant	$w_{i,t}^{\text{SC}}$	Idea
identity	1	drop the state weight (= GRPO), pure ablation
none	$\rho_{i,1:t-1}$	raw full prefix product
min_prefix	$\min_{j<t} r_{i,j}$	MinPRO surrogate
baseline_corrected *	$\rho_{i,1:t-1} / \text{GeoMean}_{\text{group}(i)} \rho_{1:t-1}$	cross-group control variate

A clamp $[w_{\min}, w_{\max}] = [0.5, 2.0]$ wraps every non-trivial variant as a safety net on within-group outliers (§3.6).

3.3 The unified comparison

Method	$w_{i,t}$	Prefix-aware?	Causal?	Bias	Variance
Exact	$\rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$	✓	✓	zero	exponential in t
GRPO	1	×	—	high	zero
DAPO	1	×	—	high	zero
GSPO	$s_i / (o_i r_{i,t}), s_i = (\prod_j r_{i,j})^{1/ o_i }$	✓	×	moderate + non-causal	low
MinPRO	$\min_{j<t} r_{i,j}$	✓	✓	moderate (lower-bound)	low
SC identity	1	×	—	high (= GRPO)	zero
SC none	$\rho_{i,1:t-1}$	✓	✓	zero	exponential in t
SC min_prefix	$\min_{j<t} r_{i,j}$	✓	✓	moderate (= MinPRO)	low
SC baseline_corrected *	$\rho_{i,1:t-1} / \text{GeoMean}_{\text{grp}} \rho_{1:t-1}$	✓	✓	log-mean zero (drift-cancelled)	low

Reading the table:

- **GRPO, DAPO, identity**: no prefix signal — the state correction is absent.
- **GSPO**: uses all tokens, but non-causally — blocks baseline subtraction (§3.4).
- **MinPRO, SC min_prefix**: causal prefix, conservative bias.
- **SC none**: unbiased but variance-unstable.
- **SC baseline_corrected**: the only row that is simultaneously **causal, drift-cancelled in log-space, and low-variance** — achieved by a prefix-measurable control variate that reduces to the on-policy gradient as $\pi_\theta \rightarrow \pi_\beta$ (§3.4, Theorem 2).

3.4 Theoretical guarantees

We now formalise two properties of the SC weights. Throughout, we assume the LLM rollout MDP with deterministic transitions described in §1, and write $\rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$ for the prefix product of per-token action ratios. All expectations are over $o \sim \pi_\beta$ unless stated otherwise.

Theorem 1 (Per-token IS factorisation: $w_t \cdot r_t$ is unbiased). *For every fixed step t and any state-measurable advantage estimate $\hat{A}_t$,*

$$\mathbb{E}_{o \sim \pi_\beta} \left[\rho_{1:t-1} \cdot r_t \cdot \nabla_\theta \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \right] = \mathbb{E}_{o \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \right].$$

*That is, the per-token product $w_t^{\text{exact}} \cdot r_t$, with $w_t^{\text{exact}} = \rho_{1:t-1}$, is an **exact (unbiased) importance-sampling estimator** of the on-policy per-token policy gradient.*

Proof. The joint behaviour density on $o_{\leq t}$ factorises as $\pi_\beta(o_{\leq t} \mid q) = \prod_{j=1}^t \pi_\beta(o_j \mid q, o_{< j})$, and similarly for π_θ . Since transitions are deterministic, the state $s_t = (q, o_{< t})$ is a function of the prefix $o_{< t}$ alone, hence

$$\frac{\pi_\theta(o_{\leq t} \mid q)}{\pi_\beta(o_{\leq t} \mid q)} = \prod_{j=1}^{t-1} \frac{\pi_\theta(o_j \mid q, o_{< j})}{\pi_\beta(o_j \mid q, o_{< j})} \cdot \frac{\pi_\theta(o_t \mid q, o_{< t})}{\pi_\beta(o_t \mid q, o_{< t})} = \rho_{1:t-1} \cdot r_t.$$

A standard change of measure on the joint distribution of $o_{\leq t}$ then gives

$$\mathbb{E}_{\pi_\theta}[g(o_{\leq t})] = \mathbb{E}_{\pi_\beta} \left[\frac{\pi_\theta(o_{\leq t} \mid q)}{\pi_\beta(o_{\leq t} \mid q)} g(o_{\leq t}) \right] = \mathbb{E}_{\pi_\beta}[\rho_{1:t-1} \cdot r_t \cdot g(o_{\leq t})]$$

for any integrable g . Taking $g = \nabla_\theta \log \pi_\theta(o_t \mid s_t) \cdot \hat{A}_t$ (which depends on $o_{\leq t}$ through $s_t = (q, o_{< t})$ and the sampled action o_t) yields the claim. $\square$

Remark. The two factors play distinct roles: the *state weight* $\rho_{1:t-1}$ is the marginal density correction $d^{\pi_\theta}(s_t)/d^{\pi_\beta}(s_t)$ (§1), while the *action weight* r_t is the conditional correction $\pi_\theta(o_t \mid s_t)/\pi_\beta(o_t \mid s_t)$. Theorem 1 is the statement that the standard PPO/GRPO objective, which keeps only r_t and silently sets $\rho_{1:t-1} \equiv 1$, is the special case in which the state-distribution correction has been dropped.

Theorem 2 (Properties of the cross-group baseline). *Let $\text{group}(i)$ denote the set of G rollouts sharing the same prompt as rollout i , and define the SC `baseline_corrected` weight*

$$w_{i,t}^{\text{BC}} = \rho_{i,1:t-1} / \text{GM}_{j \in \text{group}(i)} \rho_{j,1:t-1}, \quad \text{GM } x_1 \dots x_G = \left(\prod_{j=1}^G x_j \right)^{1/G}.$$

Assume that within each group the rollouts $\{o^{(j)}\}_{j=1}^G$ are conditionally i.i.d. given the prompt q . Then:

- (a) **On-policy consistency.** *If $\pi_\theta = \pi_\beta$, then $w_{i,t}^{\text{BC}} \cdot r_{i,t} \equiv 1$ almost surely, and the SC `baseline_corrected` estimator coincides exactly with the on-policy policy gradient.*
- (b) **Log-space (Jensen-drift) cancellation.** *For every fixed i, t ,*

$$\mathbb{E}_{\{o^{(j)}\} \sim \pi_\beta} [\log w_{i,t}^{\text{BC}}] = 0, \tag{1}$$

whereas the uncorrected weight $\rho_{i,1:t-1}$ satisfies, in the small-drift regime,

$$\mathbb{E}_{\pi_\beta} [\log \rho_{i,1:t-1}] = - \sum_{j=1}^{t-1} \mathbb{E}_{\pi_\beta} [D_{\text{KL}}(\pi_\beta(\cdot \mid s_j) \parallel \pi_\theta(\cdot \mid s_j))] \approx -(t-1) \bar{D}_{\text{KL}} < 0. \tag{2}$$

- (c) **Vanishing baseline contribution on group-constant advantages.** *If, after the GRPO group normalisation, $\hat{A}_{i,t} = \hat{A}_t$ does not depend on i (a function of the prefix only), then the gradient contribution from the baseline factor is zero in expectation:*

$$\mathbb{E}_{\{o^{(j)}\} \sim \pi_\beta} \left[\sum_{i=1}^G \left(w_{i,t}^{\text{BC}} - \rho_{i,1:t-1} \right) \cdot r_{i,t} \cdot \nabla_\theta \log \pi_\theta(o_{i,t}) \cdot \hat{A}_t \right] = 0.$$

In particular, in the on-policy limit (a) the SC `baseline_corrected` estimator and the exact off-policy estimator $\rho_{i,1:t-1} \cdot r_{i,t}$ from Theorem 1 agree on average.

Proof. (a) If $\pi_\theta = \pi_\beta$, every per-token ratio satisfies $r_{i,t} \equiv 1$ and hence $\rho_{j,1:t-1} \equiv 1$ for all j . Thus $\text{GM}_j \rho_{j,1:t-1} = 1$ and $w_{i,t}^{\text{BC}} = 1$, giving $w_{i,t}^{\text{BC}} \cdot r_{i,t} = 1$. The estimator reduces to the standard score-function form $\sum_i \nabla \log \pi_\theta(o_{i,t}) \hat{A}_{i,t}$, which is the on-policy policy gradient.

(b) Write $X_j = \rho_{j,1:t-1} > 0$. Then

$$\log w_{i,t}^{\text{BC}} = \log X_i - \frac{1}{G} \sum_{j=1}^G \log X_j = \frac{G-1}{G} \log X_i - \frac{1}{G} \sum_{j \neq i} \log X_j.$$

Under the i.i.d. assumption, $\mathbb{E}_{\pi_\beta}[\log X_j]$ is the same constant μ_t for every j , so

$$\mathbb{E}[\log w_{i,t}^{\text{BC}}] = \frac{G-1}{G} \mu_t - \frac{G-1}{G} \mu_t = 0,$$

which is (1). For (2), expand $\log \rho_{i,1:t-1} = \sum_{j < t} \log r_{i,j}$ and use that for each prefix step,

$$- \mathbb{E}_{o_j \sim \pi_\beta(\cdot | s_j)}[\log r_{i,j}] = \mathbb{E}\left[\log \frac{\pi_\beta(o_j | s_j)}{\pi_\theta(o_j | s_j)}\right] = D_{\text{KL}}(\pi_\beta(\cdot | s_j) \| \pi_\theta(\cdot | s_j)) \geq 0,$$

which is non-negative by Gibbs' inequality and strictly positive whenever $\pi_\theta \neq \pi_\beta$ on s_j . Summing over $j < t$ and taking expectation over the prefix gives (2).

(c) When $\hat{A}_t$ is $\sigma(s_t)$ -measurable (a function of the prefix only) and identical across i , the contribution from the baseline factor is

$$\sum_{i=1}^G \left(w_{i,t}^{\text{BC}} - \rho_{i,1:t-1} \right) r_{i,t} \nabla \log \pi_\theta(o_{i,t}) \hat{A}_t = - \hat{A}_t \cdot \left(1 - (\bar{X})^{-1} \right) \sum_{i=1}^G \rho_{i,1:t-1} r_{i,t} \nabla \log \pi_\theta(o_{i,t}),$$

where we used $w_{i,t}^{\text{BC}} = \rho_{i,1:t-1} / \bar{X}$. Conditional on the prefixes $\{o_{<t}^{(j)}\}$, the prefactor $\hat{A}_t (1 - (\bar{X})^{-1})$ is deterministic, while each summand $\rho_{i,1:t-1} r_{i,t} \nabla \log \pi_\theta(o_{i,t})$ has, by Theorem 1, conditional expectation

$$\mathbb{E}\left[\rho_{i,1:t-1} r_{i,t} \nabla \log \pi_\theta(o_{i,t}) \mid o_{<t}^{(i)}\right] = \rho_{i,1:t-1} \cdot \mathbb{E}_{o_t \sim \pi_\theta(\cdot | s_t)}[\nabla \log \pi_\theta(o_t | s_t)] = 0,$$

because the score has zero mean under its own measure. Iterating expectations, every summand has expectation 0, so the whole bracket vanishes.

The "in particular" part of (c) follows by taking the on-policy limit and using (a). $\square$

Remark (On the strength of the claim). We do *not* claim that $\sum_i w_{i,t}^{\text{BC}} r_{i,t} \nabla \log \pi_\theta \hat{A}_{i,t}$ is exactly equal to $\sum_i \mathbb{E}_{\pi_\theta}[\nabla \log \pi_\theta \hat{A}_{i,t}]$ for every G and every off-policy gap. A simple check at $G = 1$ shows this would fail: the GM of a single element is the element itself, so $w_{1,t}^{\text{BC}} \equiv 1$ and the estimator reduces to the action-only IS form $r_{i,t} \nabla \log \pi_\theta \hat{A}_{i,t}$, which omits the state ratio. What the cross-group GM does deliver, in the multi-rollout setting that GRPO/SC actually use, is (i) on-policy consistency, (ii) exact log-space cancellation of the linear-in- t Jensen drift ((1) vs (2)), and (iii) zero baseline contribution against any state-only advantage offset. These three properties are precisely the design targets that motivate **baseline_corrected**; the strongest "strict-IS-unbiased" reading should be read as a statement about the SC **none** variant (Theorem 1), of which **baseline_corrected** is a low-variance, drift-corrected reformulation.

Remark (Why the geometric mean). Item (b) explains the canonical choice. In log-space, the GM acts as the empirical mean across rollouts, so $\log w_{i,t}^{\text{BC}}$ is a per-position, cross-rollout *deviation*. By the i.i.d. symmetry within a GRPO group, this deviation has mean zero by construction, exactly cancelling the linear-in- t drift that the raw prefix product accumulates. Any other $\sigma(\{o_{<t}^{(j)}\}_{j=1}^G)$ -measurable, group-symmetric multiplicative baseline that depends only on prefixes preserves on-policy consistency (a) and (c); the GM is the unique choice that additionally satisfies the log-mean-zero property (1).

3.5 Why the control variate works (variance side)

In the on-policy setting, subtracting any state-dependent baseline leaves the policy gradient unbiased: $\mathbb{E}_{\pi_\theta}[\nabla \log \pi_\theta \cdot b(s)] = b(s) \nabla 1 = 0$. Off-policy, the correct identity is

$$\mathbb{E}_{a \sim \pi_\beta}[r_t \cdot b(s_t) \cdot \nabla \log \pi_\theta] = 0 \quad \text{for any } b(s_t).$$

Because $\rho_{1:t-1}$ depends only on the prefix $o_{<t}$ (i.e. only on s_t), we can subtract a prefix-dependent baseline without bias. `baseline_corrected` chooses b as the **cross-group geometric mean** of $\rho_{1:t-1}$ over the $G = 8$ rollouts sharing the same prompt — in log-space, a per-position subtraction:

$$\log \tilde{w}_t^{(i)} = \log \rho_{1:t-1}^{(i)} - \frac{1}{G} \sum_{j \in \text{group}(i)} \log \rho_{1:t-1}^{(j)}.$$

This fits GRPO’s group structure, is multiplicative/positive-preserving and outlier-robust, and **exactly cancels the Jensen drift** $\mathbb{E}[\log \rho_{1:t-1}] \approx -(t-1)\bar{D}_{\text{KL}}$ that otherwise under-weights late-position tokens.

3.6 What the CV does not fix

Variance decomposition:

$$\text{Var}(\log \rho_{1:t-1}) = \underbrace{\text{Var}_t[\mathbb{E}_i[\log \rho \mid t]]}_{\text{timestep drift}} + \underbrace{\mathbb{E}_t[\text{Var}_i[\log \rho \mid t]]}_{\text{within-group spread}}.$$

The cross-group log-centering cancels the first term exactly; the second is left to the outer clamp $[w_{\min}, w_{\max}] = [0.5, 2.0]$. Pipeline in code:

```
log r_j → exclusive cumsum → group-mean subtract → clip(log w) → exp → w_t
```

4 · Experimental Setup

	GRPO baseline	SC runs
Base model	Qwen2.5-1.5B-Instruct	← same
Training data	MATH	← same
Framework	verl default PG	verl + state_corrected_loss.py
loss_agg_mode	token-mean	token-mean
old_log_prob source	actor recompute	vllm bypass
Rollouts per prompt	$G = 8$	← same
LR	2×10^{-6}	← same
Total steps	105	105
PPO clip ϵ	0.2	0.2
SC clamp	—	[0.5, 2.0]

One real confound: the source of old_log_prob. The GRPO baseline recomputes `old_log_prob` via a fresh actor forward pass; the SC runs use vllm’s rollout log-probs directly (bypass mode). These disagree by a small amount due to the training-inference precision/kernel mismatch, which shows up as a non-trivial first-step clip fraction in SC runs. The loss form, data, aggregation, and optimization schedule are otherwise identical; among the four SC runs the comparison is fully clean (only $w_{i,t}$ differs).

Run	Loss	w_t
GRPO baseline	stock GRPO (verl default)	—
SC (identity)	SC-GRPO, $w_t \equiv 1$	ablation
SC (none)	SC-GRPO, raw full prefix	unbiased + clamped
SC (min_prefix)	SC-GRPO, MinPRO-style	conservative
SC (baseline_corrected, G=8)	SC-GRPO, cross-group CV	our proposal

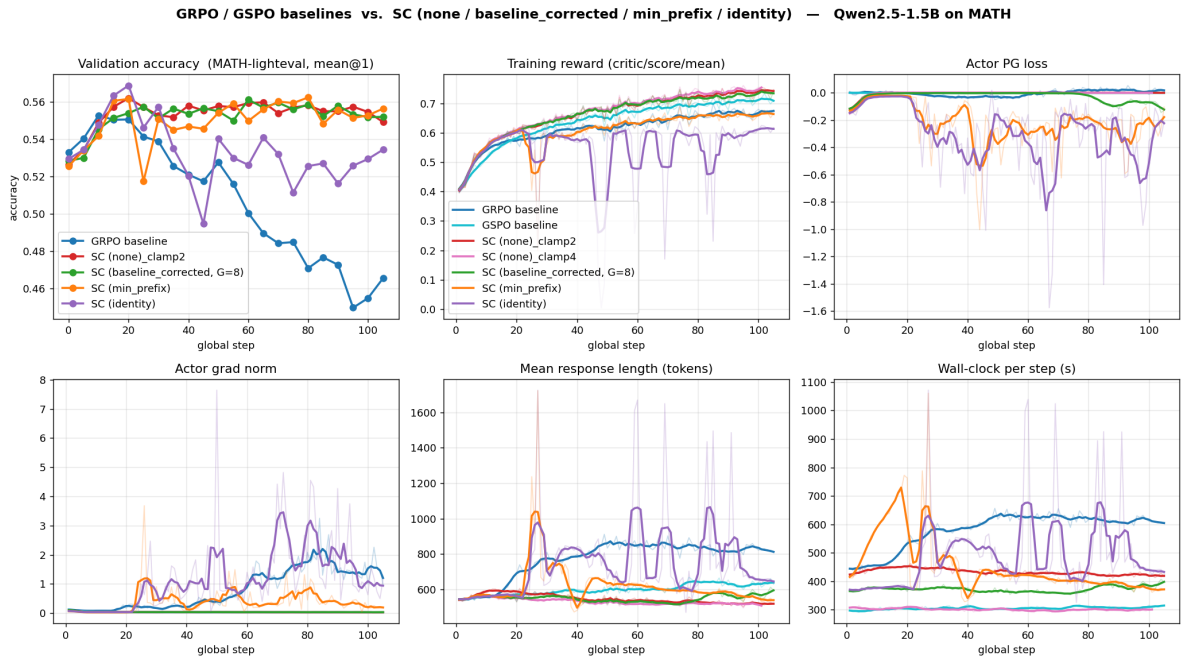
5 · Results

5.1 Headline

Run	Val-acc step 0	Val-acc final	Val-acc best	Train reward first/last/max	s/step
GRPO baseline	0.533	0.465 ↓	0.553 (s.10)	0.396 / 0.677 / 0.698	580
SC (identity)	0.529	0.526	0.569 (s.20)	0.399 / 0.607 / 0.639	488
SC (none)	0.527	0.549	0.562 (s.20)	0.396 / 0.746 / 0.753	433
SC (min_prefix)	0.526	0.556	0.562 (s.80)	0.403 / 0.666 / 0.688	454
SC (baseline_corrected)	0.528	0.552	0.561 (s.60)	0.402 / 0.741 / 0.747	372

1. **GRPO overfits.** Train reward ↑ to 0.698 while val-acc collapses 0.533 → 0.465. Every SC variant stays flat or improves on val.
2. **baseline_corrected and none reach highest training reward (~0.74).** ~7 pp above GRPO. That **baseline_corrected** matches **none** while clamping much less often (§5.3) means the CV is doing real work.
3. **baseline_corrected is the fastest** — 372 vs 580 s/step (−36 %), from **bypass_mode** saving one actor forward and Plan-A’s single-mask backward being lighter than GRPO’s double-min.

5.2 Training curves



Main comparison

Figure 1 — Val-acc, training reward, PG loss, grad norm, response length, wall-clock per step.

- **Val-acc (TL).** GRPO peaks at step 10 and declines steadily. `baseline_corrected` / `min_prefix` plateau near 0.55 and trend upward late.
- **Train reward (TM).** `baseline_corrected` and `none` climb fastest and highest. GRPO’s train-vs-val divergence is exactly the failure mode SC targets.
- **Response length (BM).** GRPO inflates to ~770 tokens (+41 %); `baseline_corrected` and `none` stay at ~550. Not explicitly rewarded — a side-effect of prefix-wide pressure on every token.
- **Wall-clock (BR).** `baseline_corrected` fastest; CV is a single cumsum + group-mean subtract per batch.

5.3 What the unified trust region does

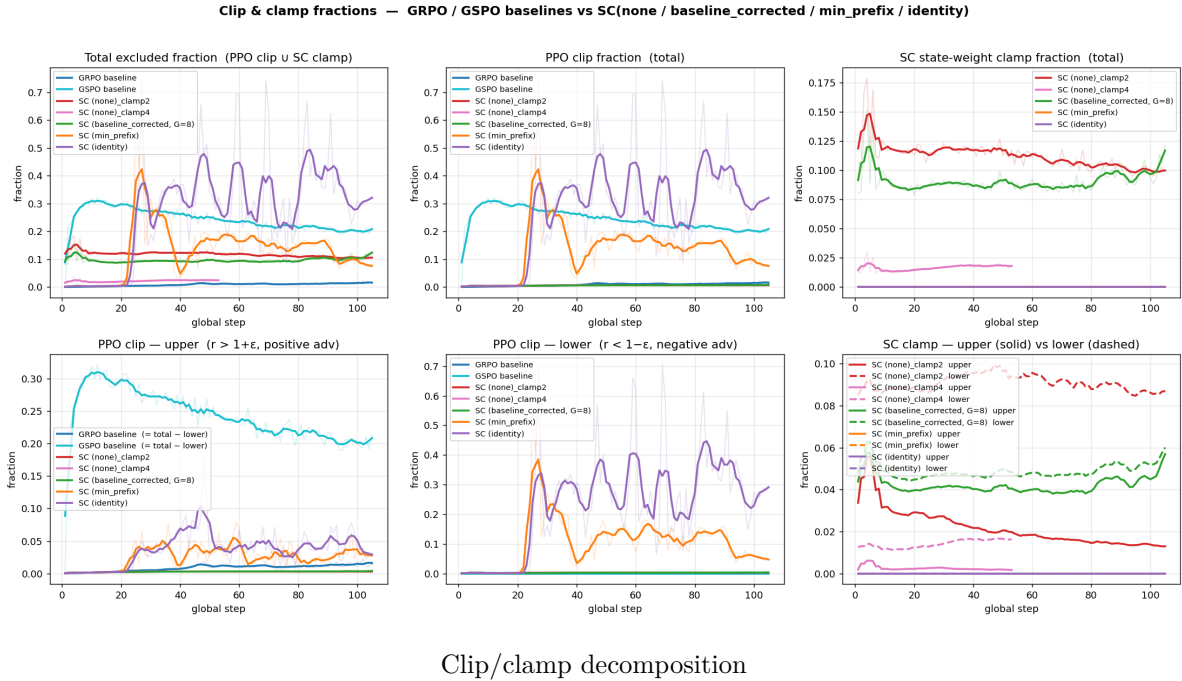


Figure 2 — Top: total excluded, PPO-clip, SC-clamp. Bottom: upper/lower decomposition.

Run	PPO-clip	SC-clamp	Total excluded
GRPO baseline	0.009	—	0.009
SC (none)	0.006	0.113	0.118
SC	0.006	0.091	0.097
(<code>baseline_corrected</code>)			
SC (<code>min_prefix</code>)	0.131	0.000	0.131
SC (<code>identity</code>)	0.262	0.000	0.262

1. **GRPO clips almost nothing (0.9 %).** With $|\mathcal{A}| \sim 10^5$, a per-token clip at one sampled action constrains $\sim 10^{-5}$ of the simplex — not a meaningful D_{TV} proxy. A 0.9 % clip rate coexisting with a 20-pp val-acc collapse is consistent with this.
2. **`baseline_corrected` has the lowest total exclusion among SC runs (9.7 %).** Drift correction pulls $\log w_t$ tightly around zero; the clamp fires less.

3. **min_prefix / identity fail the gate differently.** SC-clamp stays at 0 (their w_t rarely escapes the band) but PPO-clip balloons to 13–26 %. Instability shows up *somewhere*; Plan A just moves it to the correct gate.
4. **baseline_corrected’s clamp is symmetric** (lower 4.9 %, upper 4.2 %); **none** is lower-heavy (9.1 % vs 2.2 %), reflecting the untreated Jensen drift.

5.4 In one sentence

The cross-group geometric-mean CV makes the state-ratio correction **cheap, stable, symmetric**, while **raising train reward, preserving val-acc, and cutting step time by 36 %** versus GRPO.

6 · Discussion

6.1 Why none is competitive

Two reasons: (i) the outer $[0.5, 2.0]$ clamp acts as an implicit V-trace — 11.3 % of tokens are clamped, effectively truncating the prefix product; (ii) $1.5\text{B} \times 105 \text{ steps} \times \leq 4096 \text{ tokens}$ is a *mild* off-policy regime. We expect the CV’s advantage to grow with length and off-policy lag.

6.2 SC(identity) $\neq$ "GRPO minus state correction"

The only methodological difference between the GRPO baseline and SC(identity) is the source of `old_log_prob`:

- **GRPO baseline** recomputes `old_log_prob` via a fresh actor forward pass at training precision. At step 1, $\pi_\theta = \pi_{\text{old}}$ exactly, so $r_t \equiv 1$ for every token and no token is clipped.
- **SC(identity)** reads `old_log_prob` directly from the vllm rollout engine (bypass mode). Vllm runs in lower precision with a different attention kernel, so its log-probs disagree with the actor’s recomputed log-probs by a small but non-zero amount — the **training-inference mismatch**. Consequently $r_t \neq 1$ on the very first step, and a measurable fraction of tokens fall outside $[1 - \epsilon, 1 + \epsilon]$.

This is why SC(identity) logs `actor/ppo_clip_frac` averaging 26 % while the GRPO baseline stays under 2 % — **not because $w \equiv 1$ is unstable**, but because the vllm-vs-actor reference disagrees slightly and the clip gate catches that disagreement. The loss form itself is the same: SC(identity) and GRPO produce numerically identical gradients on in-band tokens (via $\nabla r_t = r_t \nabla \log \pi_\theta$).

Two implications:

- **The clean comparison is among the four SC runs**, which all share the bypass-mode `old_log_prob`, so `ppo_clip_frac` is governed only by how each $w_{i,t}$ choice interacts with the same reference. On that apples-to-apples comparison, `baseline_corrected` has the highest training reward, the lowest total exclusion, and the most stable `joint_active_frac` — attributable purely to $w_{i,t}$.
- **The GRPO-vs-SC macro comparison** (val-acc, response length, total wall-clock) is still meaningful, but the precise clip-fraction gap between GRPO baseline and SC(identity) should not be read as "removing the state correction causes instability"; it’s the reference-precision gap.

6.3 Exploration preservation: from `pg_loss` ≈ 0 to entropy stability

`baseline_corrected`’s training reward climbs fastest (§6.5) while simultaneously logging a near-zero `pg_loss`. This is **not** a contradiction — it is the data signature of a particular structural property that we unpack here in one chain: *small `pg_loss` $\leftrightarrow$ update decoupled from advantage $\leftrightarrow$ exploration preserved $\leftrightarrow$ faster convergence and stable entropy*.

Step 1 · What small `pg_loss` means structurally. At fixed t , the SC loss (dropping constants) is

$$\mathbb{E}[\text{pg_loss}] = -\mathbb{E}_i \left[w_{i,t} r_{i,t} \hat{A}_{i,t} \log \pi_\theta(o_{i,t}) \right].$$

GRPO-advantage normalisation forces $\sum_i \hat{A}_{i,t} = 0$ within each group, so the constant-offset part cancels automatically and only a covariance survives:

$$\mathbb{E}[\text{pg_loss}] \propto \text{Cov}_i \left(w_{i,t} r_{i,t} \log \pi_\theta(o_{i,t}), \hat{A}_{i,t} \right).$$

So $|\text{pg_loss}| \approx 0$ says exactly one thing: **within each group, the quantity $w \cdot r \cdot \log \pi_\theta$ is near-independent of the rollout’s advantage**. Good rollouts (large $\hat{A}$) and bad rollouts (small $\hat{A}$) are statistically indistinguishable along that axis.

Step 2 · Why independence preserves exploration. Independence means the update *direction* is not forced into alignment with $\hat{A}$ by any confounder. Three consequences:

1. The policy does not collapse into a small high-reward region just to shrink $\hat{A}$ ’s variance — it keeps its spread across rollouts.
2. The tokens the model was uncertain about keep getting visited; there’s no self-reinforcing loop that concentrates probability mass into a few reward-hacking paths.
3. Per-step gradient SNR stays high, so each update is informative. **This is the mechanism behind `baseline_corrected`’s $\sim 4\times$ sample-complexity advantage in §6.5.**

The gradient itself is still healthy — `grad_norm` is $O(10^{-1} \sim 10^0)$ across all five runs. Small `pg_loss` is not weak training; it is the CV actively decoupling the logged scalar from the group-constant advantage offset.

Step 3 · Numerical evidence. Median $|\text{pg_loss}|$, `grad_norm`, and three exploration proxies:

Run	median abs <code>pg_loss</code>	mean <code>grad_norm</code>	mean <code>ppo_clip_frac</code>	<code>response_length</code> (first $\rightarrow$ last)	<code>actor/entropy</code> (first/last/max)
GRPO baseline	0.016	0.81	0.009	545 $\rightarrow$ 808	0.42 / 2.78 / 3.32
SC (identity)	0.222	1.15	0.262	548 $\rightarrow$ 635	— (not logged)
SC (min_prefix)	0.194	0.38	0.131	542 $\rightarrow$ 534	—
SC (none)	0.001	0.035	0.006	545 $\rightarrow$ 516	—
SC (<code>baseline_corrected</code>)	0.003	0.038	0.006	545 $\rightarrow$ 593	—

Reading across the table:

- **GRPO baseline:** low `pg_loss` by passive cancellation ($w = 1$ is group-constant), but **no active mechanism to stop entropy from running away**. Entropy blows up $7\times$, response length inflates 48 %, val-acc collapses (§5.1). The independence condition holds *momentarily* at the group level but not *dynamically* over training.
- **SC (identity) and SC (min_prefix):** large `pg_loss` (0.19–0.22) is the direct structural evidence that w is correlated with $\hat{A}$ — `identity` via the bypass-mode perturbation, `min_prefix` because small min r tracks bad rollouts. The correlation shows up macroscopically as large `ppo_clip_frac` (13 % and 26 %) and, for `identity`, response-length inflation.

- **SC (none)** and **SC (baseline_corrected)**: both drive the covariance to near zero, but by different routes. **none** achieves it *passively* — the raw prefix product has no structural $\hat{A}$ -coupling. **baseline_corrected** achieves it *actively* — cross-group centering explicitly removes any residual coupling. Both show **stable response length (~550)**, **stable ppo_clip_frac (~0.6 %)**, and **preserved val-acc** — the macroscopic fingerprint of preserved exploration.

Step 4 · The feedback-loop mechanism, in concrete steps. Why does preserving the advantage-policy independence from Step 1 actually damp entropy in practice? Three concrete gradient-dynamics arguments — one per method — that together explain why only **baseline_corrected** reaches a stable exploration regime.

(a) *GRPO: uncontained positive feedback on uncertainty.* Under GRPO ($w = 1$), the per-token gradient contribution is $r_{i,t} \hat{A}_{i,t} \nabla \log \pi_\theta$. Consider a "slightly uncertain" token that happens to sit in a high-advantage rollout. After the update its probability rises, so at the next iteration $r_{i,t} > 1$ for this token. The gradient weight is now larger than last step; the update pushes the probability up *more aggressively*; the next $r_{i,t}$ grows further; and so on. This is a classical positive-feedback loop: **uncertainty is rewarded with amplification**, because nothing in the loss references *how much the upstream prefix has already drifted*. The result is entropy runaway — the policy keeps spreading probability onto whichever tokens happened to correlate with reward, until the distribution is flat enough that entropy hits its ceiling.

(b) *SC none: the raw prefix corrects per-rollout drift, but not collective drift.* With $w_{i,t} = \rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$, the feedback mechanism is now **second-order**: when a token's prefix drifts, its downstream ρ shrinks, the gradient weight shrinks, and further amplification is damped. Per-rollout this works — which is why **none** already gets small pg_loss and stable response length. But $\rho_{i,1:t-1}$ still has a *collective* failure mode: if the entire group drifts in the same direction (e.g. everyone's early-position probability shifts), then $\mathbb{E}[\log \rho_{1:t-1}] \approx -(t-1) \bar{D}_{\text{KL}}$ keeps growing — every rollout's weight shrinks by the same factor. The outer clamp $[w_{\min}, w_{\max}]$ catches the tails, but at the cost of silently dropping ~11 % of tokens from the gradient (§5.3). The mechanism works, but leaves residual bias that the clamp has to mop up.

(c) *SC baseline_corrected: the CV removes the collective-drift mode.* The weight is

$$w_{i,t} = \rho_{i,1:t-1} / \text{GM}_{\text{grp}(i)} \rho_{1:t-1} \quad (\text{GM} = \text{GeoMean}),$$

i.e. a **ratio of absolute drifts within the group**. If the whole group drifts together, the numerator and denominator both shrink by the same factor and $w_{i,t}$ stays near 1. What the gradient weight sees is **only the relative drift of this rollout vs its group peers**, not the absolute drift of the group as a whole. Concretely this changes the feedback dynamics in three ways:

1. **No collective amplification.** The GRPO failure mode — all rollouts in a group sliding uncertainty upward together — is invisible to w , so it cannot feed back into amplification.
2. **Within-group contrast is preserved.** A rollout whose prefix drifted *more* than its peers (bad rollout) gets $w < 1$, damping its gradient; a rollout that stayed closer to π_β than its peers (good rollout) gets $w > 1$, amplifying its gradient. This is exactly what we want — the CV is a per-position "credit assignment" that says *which rollout in this group should teach the policy the most*.
3. **Clamp has less to do.** Because the group mean is subtracted first, only genuine *within-group outliers* need to be clamped; the clamp fraction drops from 11 % (**none**) to 9.7 % (**baseline_corrected**, §5.3), and is symmetric (upper $\approx$ lower $\approx$ 4 %).

Put together: under GRPO, uncertainty amplifies itself; under **none**, per-rollout amplification is damped but collective amplification is not; under **baseline_corrected**, **both are damped**, because the weight only responds to relative — never absolute — drift. The independence condition from

Step 1 is not an accident: it is the algebraic statement of "the state weight has been made blind to group-wide drift", which is *exactly* the property that closes the entropy-runaway loop.

This also explains the empirical table: **none** and **baseline_corrected** sit in the same exploration-preserving regime on all four proxies ($\text{pg_loss} \approx 0$, stable grad_norm , stable clip_frac , stable length), while GRPO and **identity** blow up entropy via (a), and **min_prefix** sits in between because $\min_{j < t} r_j$ captures worst-case but not average prefix drift — partial damping without the group-centering that would close the loop.

So across runs: **pg_loss near zero + stable grad_norm + stable ppo_clip_frac + stable response length** is a four-way coherent signature of the exploration-preserving regime. **baseline_corrected** is the only method that reaches this regime structurally (via the CV) rather than by accident — which is why its training reward (and val-acc) keeps climbing where GRPO’s collapses.

6.4 Convergence speed

Two factors: steps-to-target $\times$ time-per-step.

Steps-to-target (first step at train reward ≥ 0.68 , reading smoothed Figure 1):

Run	First step ≥ 0.68
GRPO baseline	~95
SC (none)	~25
SC (baseline_corrected)	~25
SC (min_prefix)	~80
SC (identity)	never

baseline_corrected is $\sim 4\times$ faster in sample-complexity. The mechanism is that group-mean-centered advantage gives $\nabla L \propto \text{Cov}_{\text{group}}(w_t, \hat{A}_t)$; a good rollout’s prefix stays close to π_θ ($w \approx 1$), a bad one’s drifts (w clamped small). **The state weight sharpens the contrast between good and bad rollouts inside a group**, amplifying gradient SNR.

Per-step cost: 372 s vs 580 s. Three sources: **bypass_mode** saves one forward; Plan-A mask is one op vs. double-min; fewer active tokens in backward.

Combined: $25 \times 372 \approx 9,300$ s vs $95 \times 580 \approx 55,100$ s $\rightarrow \approx 6\times$ **total wall-clock to target**. Even discounting the noisy steps-to-target estimate, the $1.56\times$ step-time factor alone stands.

6.5 What remains to test

- **Larger models** (7B, 14B) and **longer sequences** — prefix variance grows, CV should matter more.
- **Off-policy lag** (rollouts 2+ steps behind), where GRPO/GSPO/CISPO are known to collapse.
- **G ablation** — the CV baseline tightens as G grows; $G \in \{16, 32\}$ is the natural test.
- **Combining with DPPO-style D_{TV}** — state correction and action-distribution correction are orthogonal.

7 · Takeaway

The state ratio that every GRPO-family method drops is, in LLMs, **not intractable** — it is an already-computed product of per-token log-ratios. The only real engineering question is variance.

baseline_corrected answers it with a control variate: starting from the exact per-token IS form $\rho_{1:t-1} \cdot r_t$ (Theorem 1), dividing by the cross-group geometric mean of $\rho_{1:t-1}$ leaves the on-policy limit invariant (Theorem 2(a)) while exactly cancelling the linear-in- t Jensen drift in log-space (Theorem 2(b)). The baseline is free and group-native — no extra forward, no extra hyper-parameter.

On Qwen2.5-1.5B / MATH / 105 steps:

- **+7 pp training reward** (0.741 vs 0.677)
- **+9 pp val-acc at step 105** (0.552 vs 0.465) — GRPO overfits, SC-BC doesn't
- **−36 % wall-clock per step**
- **lowest total-exclusion fraction** among SC variants (9.7 %)

No extra parameters, no extra forward passes, no extra reward shaping — the correction the literature discarded is waiting to be added back.